Smart Light Monitor

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1 Project Description

The goal of this project was to develop a light monitoring system using Arduino Uno. The system measures ambient light intensity using a photoresistor and displays the result as a percentage from 0 to 100 on a 16x2 LCD screen.

Three buttons are used to control the system. Two buttons increase or decrease the light threshold in steps of 5%, while the third button enables or disables the alarm. When the light level falls below the selected threshold, a piezo buzzer produces an audible warning.

The system uses a hardware timer, an external interrupt, and EEPROM to store the light threshold and alarm state even after power is disconnected.

Parts used:

- Arduino Uno
- Photoresistor (LDR) and 2 kΩ resistor
- 16x2 LCD display
- 3 push buttons
- Piezo buzzer
- Breadboard and jumper wires

2 Design and Timing Budget

The system uses an event-driven design without blocking `delay()` calls. Timer1 is configured in CTC mode with a prescaler of 64 and generates an interrupt every 1 ms. The interrupt increments a counter used to schedule tasks in the main loop.

The photoresistor is sampled every 200 ms, the LCD is updated every 200 ms, and buttons are checked every 15 ms. The buzzer operates in a cycle of 300 ms ON and 700 ms OFF.

The ON/OFF button uses an external interrupt on pin D2. The interrupt sets a flag, which is processed in the main loop after a short debounce check.

EEPROM stores the light threshold and alarm state. Magic and version values are used to validate stored data, and unchanged EEPROM bytes are not rewritten.

3 Test Results and Known Issues

The system was tested using a physical Arduino Uno. The LCD displayed light readings, and the measured value changed when the photoresistor was covered or exposed to light. All three buttons worked, allowing the threshold and alarm state to be adjusted.

The piezo buzzer activated when the measured light level fell below the configured threshold and stopped when the alarm was disabled or sufficient light was detected.

The light percentage is based on the Arduino analog reading rather than calibrated lux measurements. Therefore, the displayed value may not reach 100% under normal room lighting.

Possible improvements include calibrating the light sensor.